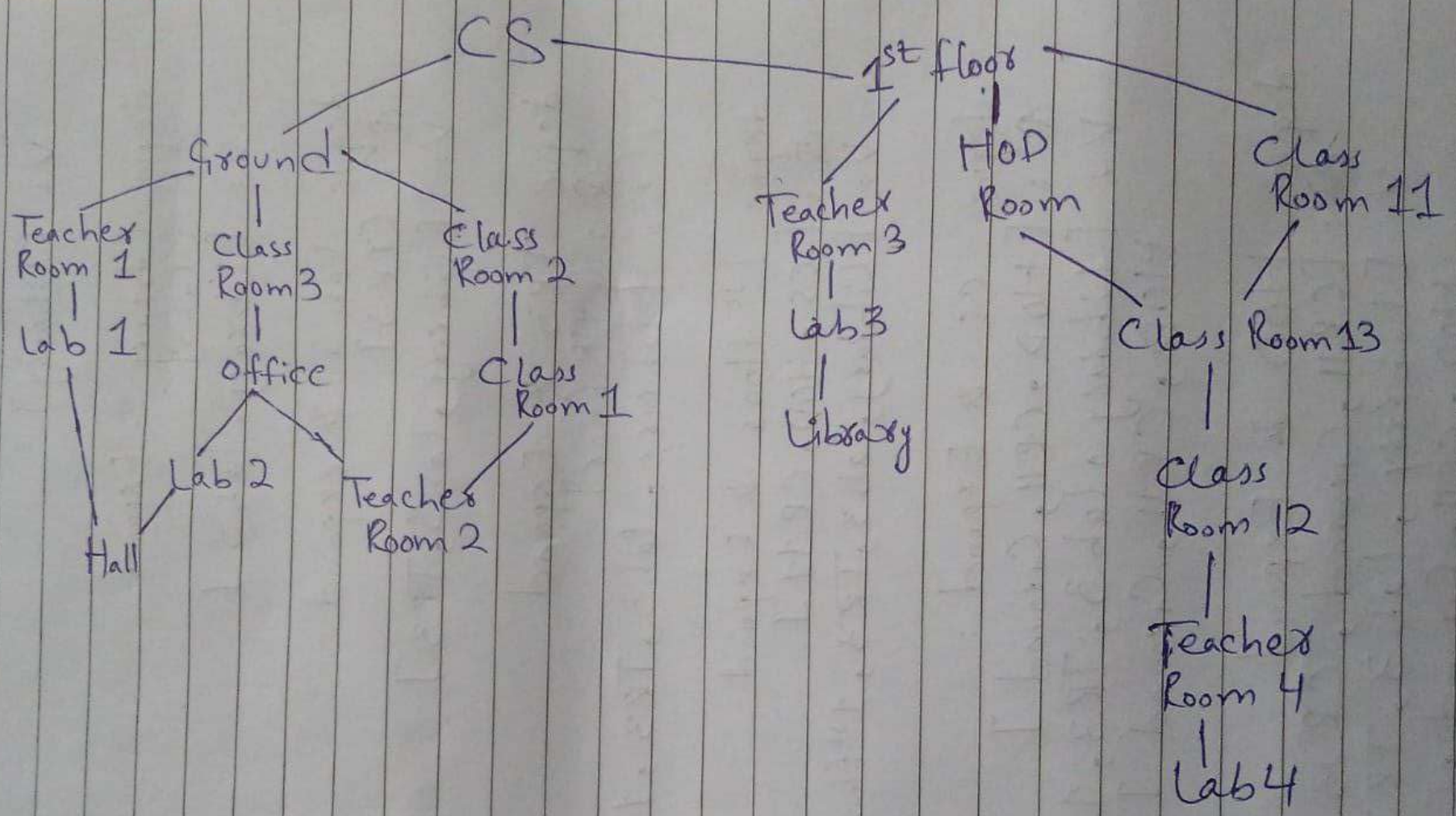
Date

GRAPH


CS
Ground
1st floor
Teacher Room 1
Class Room 3
Class Room 2
Teacher Room 3
HOD Room
Class Room 11
Lab 1
Office
Class Room 1
Lab 3
Class Room 13
Lab 2
Teacher Room 2
Library
Class Room 12
Hall
Teacher Room 4
Lab 4

Date:

TRANSITION TABLE :

CS	→	Ground , 1st floor
Ground	→	Teacherroom 1, Classroom 3, Classroom 2, CS
Teacher Room 1	→	Ground , Lab 1
Classroom 3	→	Ground, Office
Classroom 2	→	Ground, classroom 1
Lab 1	→	Teacherroom 1 , Hall
Office	→	Classroom 3, Lab 2 , Teacher room 2
Classroom 1	→	Classroom 2, Teacherroom 2
Lab 2	→	Office , Hall
Teacher room 2	→	Office , Classroom 1
Hall	→	Lab 1 , Lab 2
1st floor	→	Teacherroom 3, HODroom , Classroom 11
Teacherroom 3	→	1st floor , Lab 3
HOD room	→	1st floor , Classroom 13
Classroom 11	→	1st floor , classroom 13
Lab 3	→	Teacher room 3 , Library
Classroom 13	→	HoDroom, Classroom 13, Classroom 11
Classroom 12	→	Classroom 13, Teacher room 4
Teacher room 4	→	Classroom 12 , Lab 4
Lab 4	→	Teacher room 4
Library	→	Lab 3

Date:

Formulation of the Problem:

Initial State:
Library

Goal State:
Classroom 13

State space:
All accessible location in the CS department.

Action:
Move from one accessible state to an other adjacent state.

Transition model:
Purpose to move state

Path cost:
Distance.

BFS TABLE

Date

Initialization:
Q: [library]
V: [library]

deque Library & Expand
Q: [lab3]
V: [library, lab3]

deque lab3 & expand
Q: [TR3]
V: [library, lab3, TR3]

deque TR3 & Expand
Q: [1st floor]
V: [library, lab3, TR3, 1st floor]

deque 1st floor & Expand
Q: [CS, HOD, CR11]
V: [library, lab3, TR3, 1st floor, CS, HOD, CR11]

deque CS & Expand:
Q: [HOD, CR11, Ground]
V: [library, lab3, TR3, 1st floor, CS, HOD, CR11, Ground]

deque HOD & Expand
Q: [CR11, Ground, CR13]
V: [library, lab3, TR3, 1st floor, CS, HOD, CR11, Ground, CR13]

RC

Signature

SHORTEST PATH (Library - CR 13)

Date

Library.

↓

Lab 3

↓

Teacher Room 3

↓

1st floor

↓

HOD Room

↓

Class Room 13